

Lynbrook Math Summer Camp 2026

G1 Angle Chasing

Lynbrook Math Club

August 2026

G1.1 Lecture

What is Angle Chasing?

This method involves deducing unknown angle measures by using established geometric rules and known data from a figure. By tracing how various angles interact, hidden properties within a diagram can be revealed.

Fundamental Definitions

Before tackling more advanced configurations, we must first define our vocabulary.

Definition G1.1.1 (Angle Types). Whenever two rays or lines meet, an angle is created. We generally measure these in degrees, where a complete revolution is 360° . Angles fall into the following categories:

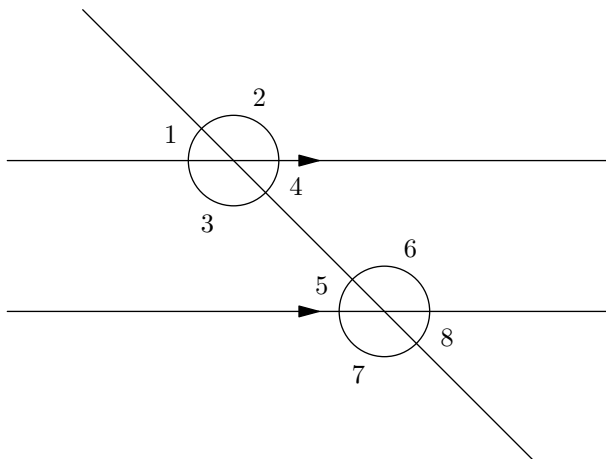
- **Acute:** greater than 0° but strictly less than 90° .
- **Right:** precisely 90° .
- **Obtuse:** greater than 90° but strictly less than 180° .
- **Straight:** perfectly flat at exactly 180° .

Definition G1.1.2 (Angle Pairs). A pair of angles is called **complementary** if their measures add up to 90° , and **supplementary** if their combined measure totals exactly 180° .

Parallel and Perpendicular Lines

Definition G1.1.3 (Line Relationships). In a given plane, two lines ℓ_1 and ℓ_2 are deemed **parallel** (written as $\ell_1 \parallel \ell_2$) if they never cross. Conversely, they are **perpendicular** (written as $\ell_1 \perp \ell_2$) if their intersection forms a perfect 90° angle.

Whenever a transversal line slices through a pair of parallel lines, it generates multiple sets of congruent angles.


Theorem G1.1.4 (Alternate Interior & Vertical Angles)

If a transversal cuts across parallel lines, the alternate interior angles produced are equal in measure. Furthermore, any two intersecting lines will universally produce congruent vertical angles.

Referring to the schematic above, we can observe that $\angle 1 \cong \angle 4 \cong \angle 5 \cong \angle 8$ alongside $\angle 2 \cong \angle 3 \cong \angle 6 \cong \angle 7$. Because of this property, you can demonstrate that two lines are parallel (if it isn't stated in the problem) simply by proving that alternate interior or corresponding angles match (for instance, showing $\angle 4 \cong \angle 5$ or $\angle 3 \cong \angle 6$).

Triangles and Polygons

Definition G1.1.5 (Triangle Classifications). We can classify triangles based on either their interior angles or their side lengths:

- **Acute:** all interior angles are strictly less than 90° .
- **Right:** features exactly one 90° interior angle.
- **Obtuse:** contains a single angle strictly greater than 90° .
- **Equilateral:** possesses three identical side lengths (and subsequently, three identical angles).
- **Isosceles:** has at least two equal side lengths (or two equal angles).
- **Scalene:** features three distinct side lengths and angles.

Definition G1.1.6 (Polygon Classifications). A polygon is a flat, enclosed geometric figure constructed from straight, non-overlapping segments. We classify a polygon as:

- **Convex:** having all interior angles under 180° . Alternatively, a segment connecting any two interior points will never cross outside the shape's boundary.
- **Concave:** any polygon that fails the convex condition.
- **Equiangular:** having all internal angles exactly equal.
- **Equilateral:** having all edge lengths equal.
- **Regular:** a shape that is simultaneously equilateral and equiangular.

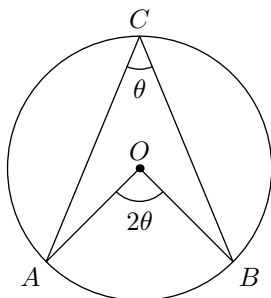
Theorem G1.1.7 (Polygon Angle Sums)

The interior angles of any triangle will invariably sum to 180° . Extending this logic, the internal angles of any n -sided polygon will always total $180^\circ(n - 2)$.

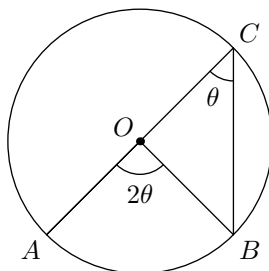
We can readily prove the triangle angle sum rule by constructing a line through one vertex that runs parallel to the opposing base, then utilizing the alternate interior angle theorem we discussed previously to "move" the base angles to the parallel line, forming a straight 180° line.

Angles in Circles

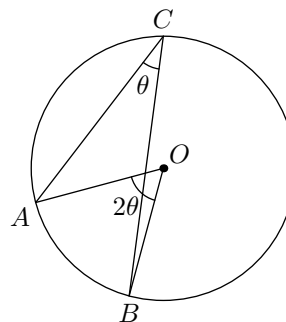
Angles interacting with circles can initially appear tricky, but the inscribed angle theorem simplifies things significantly. This rule dictates that an inscribed angle's measure is exactly half of the arc it intercepts. Consequently, for a circle centered at O with points A , B , and C on its boundary, $\angle AOB = 2\angle ACB$ (because the central angle $\angle AOB$ is equal to the measure of arc AB).



Inside



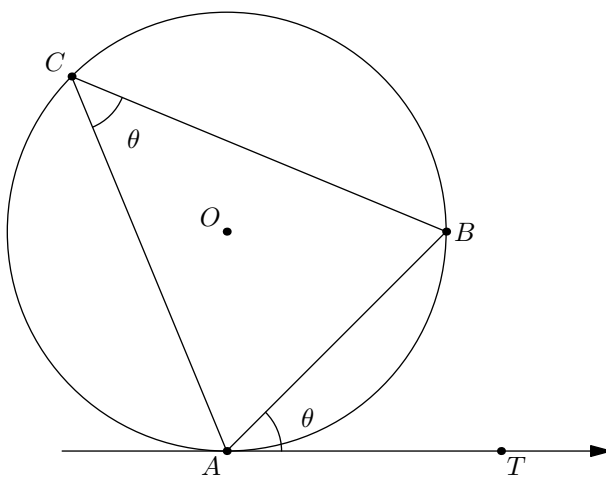
On side



Outside

As a helpful corollary, note that the triangles formed with the center, namely $\triangle AOB$, $\triangle BOC$, and $\triangle COA$, are always isosceles since their sides are the circle's radii.

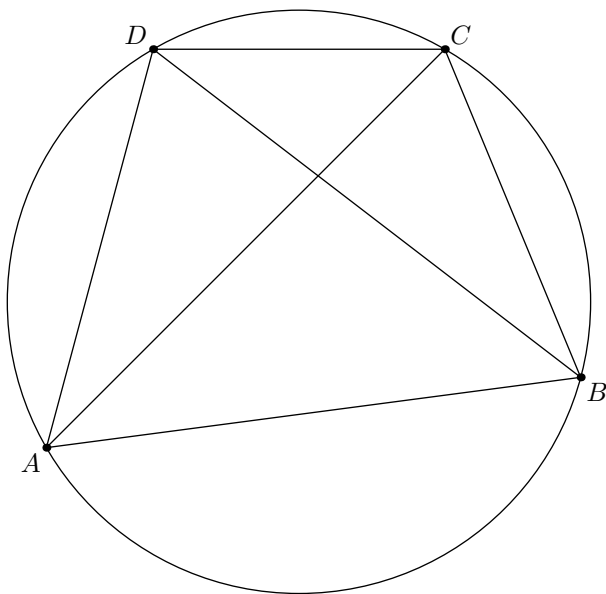
There is a highly useful alternate form of the inscribed angle theorem involving tangency. Suppose O remains the circle's center, with A , B , and C on the perimeter, and let line segment $\overline{AT}$ be tangent to the circle at A . Under these conditions, the angle formed by the tangent and the chord equals the inscribed angle: $\angle BAT = \angle BCA = \theta$, and naturally, $\angle BOA = 2\theta$.



Cyclic Quadrilaterals

Finally, we will look into cyclic quadrilaterals. These configurations frequently arise in both standard computational exercises and advanced olympiad geometry, making them crucial to spot.

A quadrilateral $ABCD$ is defined as cyclic if a single circle can be drawn that perfectly intersects all four of its vertices (A , B , C , and D).



Any of the following criteria is sufficient to prove that quadrilateral $ABCD$ is cyclic:

- Opposite angles $\angle ABC$ and $\angle ADC$ sum to 180° (they are supplementary).
- Opposite angles $\angle BCD$ and $\angle BAD$ sum to 180° (they are supplementary).
- Angles subtending the same segment are equal (e.g., $\angle ADB = \angle ACB$).
- $\angle DCA$ and $\angle DBA$ have the same measure (subtending segment AD).
- $\angle CBD$ and $\angle CAD$ have the same measure (subtending segment CD).
- $\angle BAC$ and $\angle BDC$ have the same measure (subtending segment BC).